

REVENG

DE1 Materials & Manufacturing: REVENG Report

Russell Hobbs Food Collection Hand Mixer



Figure 1: Russell Hobbs Hand Mixer [1]

Krusty Kompany

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Technical Summary

This report analyses the materials and manufacturing processes of the Russell Hobbs Hand Mixer to explore redesign alternatives that align with the Krusty Kompany's goal of creating a safer, child-friendly, and playful product.

Firstly, a background analysis of the mixer and the selected competitors identified key design differences. This was followed by a product teardown documenting the quantity, material, mass, manufacturing method, and environmental impact of each component in terms of embodied energy and CO₂ footprint. The analysis revealed that material selection contributed the most significant impact, 54% of total energy use and 51% of CO₂ emissions. ABS was the main contributor, accounting for 22% of the embodied energy and 17% of the CO₂ emissions due to its high processing energy and large mass (27% of the product). The second largest impact was from the use of the product, where the motor represented 31% of the total energy consumption and 28% of the CO₂ emissions.

The original design prioritized cost efficiency, using cheap materials and a budget motor, leading to overheating issues. The redesign improved safety, ergonomics and playfulness by replacing the ABS casing with polypropylene. This change provides a softer feel (Young's modulus: 0.824–1.02 GPa, Vickers hardness: 8 HV), lower weight (density: 895-909 kg / m³ vs ABS 1030-1060 kg / m³) and it would deform instead of shatter when impacted (elongation: 112–483% vs. ABS 5–60%). Overmolded butyl rubber grips were added for thermal insulation and enhanced ergonomics, improving comfort and grip even with wet or greasy hands. The 125 W brushed motor was replaced with a 75 W brushless motor, which is 90% energy efficient and generates less heat. The lower wattage also prevents the mixer from being too powerful for child users, which addresses a common complaint about the original design.

An environmental impact reassessment confirmed that these changes reduced the embodied energy by 18% and CO₂ emissions by 14%, primarily due to the lower motor wattage and the reduced energy required for polypropylene injection moulding. This redesign successfully aligns with Krusty Kompany's objectives while improving safety, usability, and environmental performance.

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1 Introduction

This report analyses the Russell Hobbs Food Collection Hand Mixer [2], focusing on product teardown, material evaluation, and manufacturing processes to inform design improvements. The objective was to develop solutions that align with Krusty Kompany's emphasis on safe, playful, kid-friendly designs. The environmental impact must also be considered when designing anything.

To achieve this, extensive desk research was conducted, followed by a layer-by-layer disassembly of the product to document its components and materials. A detailed Bill of Materials was compiled, and the *Eco Audit Tool* within Granta Edupack 2022[3] was utilized to assess the environmental impact of each material, specifically analysing embodied energy and CO₂ footprint. Based on these findings, a redesign strategy was proposed using reverse engineering principles. The goal was to identify alternative materials and manufacturing methods that maintain the core functionality of the product while better aligning with the client's goals.

2 Background

2.1 General Information

Generally a hand mixer is a small kitchen appliance that is used to mix, whip, and beat ingredients. The Russell Hobbs Food Collection Hand Mixer was chosen for discussion. It has a motorized base with six speeds and detachable beaters, which are mainly used to mix egg whites, cake mixture, and batter.

- **Main Features:** The mixer consists of a hard white outer shell made of ABS with two chrome-plated rotating low-carbon steel beaters. It features a 125 W motor and a speed controller. The total weight is 0.77 kg [1]
- **Retail Cost:** The Russell Hobbs website lists the product at £17.99 [2] making it a budget-friendly option.
- **Target Users:** Primarily designed for adult users who require a compact, efficient, and cheaper appliance to perform mixing tasks.

Customer reviews describe the hand mixer as a good value for money, affordable, with easy-to-understand features, and a compact, lightweight design ideal for small amounts. However, some users reported overheating, burning smells, and a fast start on the first speed setting [1].

2.2 Comparing against Pre-Existing Products



(a) Cheap Hand Mixer[4]



(b) High-Quality Hand Mixer[5]



(c) Child-Friendly Hand Mixer[6]

Figure 2: Comparison of Different Hand Mixers

Russell Hobbs Hand Mixer is budget-friendly, but not the cheapest. The £10.92 model, shown in Figure 2a is 60 g lighter, has five speeds instead of six, and a more powerful 150 W motor. However, both experience overheating and potential burning smells [7], with no major design differences. In contrast, higher-quality mixers, such as the one in Figure 2b, generally feature more powerful motors, with this model having 250 W. Its ergonomic handle and die-cast aluminium casing (510 MPa tensile strength, 650 °C melting point[3]) provide superior durability and heat resistance compared to ABS plastic (34.5 MPa tensile strength, 102°C glass transition [3]) used in budget models. Although die-cast aluminium is ideal for heavy duty mixing, ABS makes the Russell Hobbs Hand Mixer 380 g lighter, offering a more cost-effective and lightweight design.

Comparing the Russell Hobbs Hand Mixer to a more child-friendly alternative reveals significant design differences. The child-friendly model, in Figure 2c operates using a hand crank mechanism instead of an electric motor, making it safer for children by eliminating electrical hazards and the risk of overheating seen in the Russell Hobbs mixer. Furthermore, the absence of a motor and electrical components significantly reduces its weight by 540 g [8]. This lightweight aspect likely makes it very easy for children to use. The design further enhances usability for children with a non-slip grip for improved control.

3 Method

3.1 Device Teardown

The teardown process involved disassembling the Russell Hobbs Hand Mixer to analyse its components, materials, and manufacturing methods. The procedure was carried out on a layer-by-layer approach, with each component carefully removed, documented, and weighed. A complete teardown of the Russell Hobbs Hand Mixer is shown in Figure 6 in Appendix A.

- **Layer 1:** The teardown began with the removal of the beaters from the mounting mechanism. The two outer shell halves were unscrewed and carefully removed, revealing the internal components. The power cable, speed control mechanism, and vent covers were removed sequentially.
- **Layer 2:** With the casing removed, the motor, gears, springs, and cooling fans were extracted. The screws securing the motor housing were unscrewed, allowing separation of the individual parts. The electrical wiring and connections connecting the speed controller and mains cable to the motor were cut.
- **Layer 3:** The motor was further disassembled, including the rotor, stator, housing, and graphite contacts. The internal wiring and capacitors were detached from the body and documented.

3.2 Material Identification

During the disassembly of the Russell Hobbs Hand Mixer, the material for each component needed to be identified. Although some components clearly indicated their material, most required further analysis.

To determine the material, educated estimates were made based on its attributes. As most of the construction was plastic, a series of plastic identification tests were carried out. To begin, a hot soldering iron was used on the sample to assess its material type: if it softened, it was classified as thermoplastic; if it remained rigid, it was identified as a thermoset. The sample was then placed in water to check for buoyancy. Lastly, a flame test was conducted, which noted whether the material burned, the colour of the flame, the characteristics of the smoke, the odour, the dripping behaviour and the burning speed. These observations were compared with a reference flow chart, shown in Figure 9 in Appendix D.

4 Analysis

4.1 Bill of Materials

Once the material identification process was complete, a bill of materials was prepared to record all the components of the Russell Hobbs Hand Mixer. The complete bill of materials can be found in Table 3 in Appendix C. The majority of components are made from plastics such as ABS, PVC, PE, and PTFE, with the internals being low-carbon steel, and zinc. The primary manufacturing methods are injection moulding for plastics and stamping, thread rolling, and die casting for metal parts.

Table 1 shows the parts that were focussed on for improvements based on their materials and their manufacturing methods. A confidence level was defined to show the certainty of the material and the manufacturing process for each component.

- **C:** Confirmed; by testing or observing the markings.
- **EE:** Educated estimate; from component properties.

- **G:** Guess; based on limited information.

Table 1: Bill of Materials for Focused Components

Part No.	Quantity	Name	Material	Mass (kg)	Manufacturing Method	Embodied Energy (MJ)	CO ₂ Footprint (kg)
4	1	Shell (Upper)	ABS (C)	0.106	Injection Moulding (C)	12.10	0.54
5	1	Shell (Lower)	ABS (C)	0.084	Injection Moulding (C)	9.59	0.43
6	1	Control Panel	ABS (C)	0.015	Injection Moulding (C)	1.71	0.08
25	1	Rotor	Low Carbon Steel (EE)	0.111	Stamping and Winding (C)	4.52	0.37
27	1	Rotor Bracket	Zinc (EE)	0.025	Die Casting (C)	1.43	0.11
28	2	Graphite Contacts	Graphite (C)	0.001	Moulding (EE)	0.08	0.01
29	1	Stator	Low Carbon Steel (EE)	0.133	Stamping (C)	5.42	0.44

The Russell Hobbs Hand Mixer has a unique feature; the casing is secured with deeply embedded Torx security screws [9], which require specialized tools for disassembly. These screws are uncommon in household toolkits and resist tampering, making them difficult for children to remove without the proper driver. Their deep placement also prevents them from prying with objects such as coins or screwdrivers. This design already prevents children from accessing internal components, avoiding potential injuries from the electrical parts. Given that this safety measure is already in place, the redesign can focus on other aspects of child-friendly improvements.

4.2 Eco Audit

Granta EduPack[3] was used to carry out an Eco-Audit of the Russell Hobbs Hand Mixer, analysing the embodied energy and CO₂ footprint across materials, manufacturing, transport, usage, and end-of-life. This assessment provides an overall estimate as certain assumptions were made and there is the possibility of inaccuracies in material identification.

4.2.1 Materials Impact

Material selection significantly impacts the environmental footprint. ABS had the highest impact, contributing 22% of embodied energy and 17% of CO₂ emissions, due to its high processing energy and mass. To reduce this, materials with lower processing energy and embodied emissions should be considered for the main casing, minimizing the overall environmental impact.

4.2.2 Manufacturing Impact

The manufacturing method had a smaller overall environmental impact compared to material selection. Among the processes, injection moulding had the greatest impact, being responsible for 48% of the embodied energy and 42% of the CO₂ emissions, mainly due to high melting temperatures and material waste. Stamping, used for metal components such as the rotor and stator, contributed 21% to embodied energy and 19% to CO₂ emissions. Die casting, mainly for zinc parts, followed by 12% of embodied energy and 11% of CO₂ emissions.

4.2.3 Transport Impact

Transport had minimal impact on the overall footprint. It was assumed to involve sea freight (20,000 km) from Russell Hobbs factories in China to the UK, followed by 300 km of road freight. The sea freight from China to the UK accounted for 92.7% of the transport emissions, with the remainder from road freight. A potential improvement is local sourcing and manufacturing, which would reduce transport emissions but would likely increase cost.

4.2.4 Usage Impact

Usage had the second largest impact, accounting for 31.3% of energy consumption and 28.3% of CO₂ emissions. This is primarily due to the 125 W motor. The use was estimated at 10 minutes per day for 40 days per year, with a total lifespan of 5 years. Although the motor was not particularly powerful, over the 5-year lifespan, this

energy accumulates significantly. Lowering motor power or switching to a brushless motor could reduce energy consumption and CO₂ emissions. Brushless motors are 90% efficient, compared to 80% for brushed motors[10], reducing energy loss through friction and heat, and minimizing the risk of overheating.

4.2.5 End-Of-Life

The products end-of-life was landfill because all components were considered non-recyclable for the regular consumer due to the difficulty of disassembling the product. This was ignored in the overall impact breakdown table.

4.2.6 Overall Impact

A breakdown of the overall impact was the following:

Table 2: Eco-Audit Results for the Russell Hobbs Hand Mixer

Phase	Energy (MJ)	Energy (%)	CO ₂ Footprint (kg)	CO ₂ Footprint (%)
Material	54.75	54.48	2.86	50.95
Manufacture	10.78	10.73	0.91	16.19
Transport	3.37	3.36	0.24	4.33
Use	31.41	31.26	1.59	28.31
Total (First Life)	100.49	100	5.61	100

A graph generated by the Eco-Audit tool that illustrates the embodied energy and CO₂ for each phase is shown in Figure 10 in Appendix E. A graph visualising the environmental impact across different materials used in the Russell Hobbs Hand Mixer is shown in Figure 7 and 8 in Appendix B.

4.3 Materials and Manufacturing Discussion

This section analyses the materials and manufacturing processes used to create the Russell Hobbs Hand Mixer, all material properties and manufacturing data have been sourced from Granta Edupack. [3].

4.3.1 Main Casing (ABS)

The majority of the casing of the hand mixer is made of ABS, with only a few smaller components differing. The casing shell has a shiny and glossy finish on the outside surface and a rough unfinished inside. ABS contributed 206 g to the total mass of the hand mixer.

Advantages

- **Impact Resistance:** ABS has excellent impact resistance, making it durable and capable of withstanding rough handling and drops. The fracture toughness ranges from 1.46 - 4.29 MPa·m^½.
- **Lightweight:** With a density range of 1,030 – 1,060 kg·m⁻³, ABS is lightweight, making it more suitable for children to handle.
- **Aesthetic Versatility:** ABS takes colour easily and can achieve glossy finishes or vibrant textures, allowing customization for playful designs that would appeal to children.

Disadvantages

- **Ergonomics:** ABS has a rigid and smooth surface, which makes it uncomfortable to hold for extended periods. Its lack of texture reduces grip, increasing the likelihood of slipping during use.
- **Thermal Limitations:** Not suitable for applications with prolonged heat exposure, such as from the motor, due to a maximum service temperature range of 62.9 - 76.9 °C. This could cause warping or loss of mechanical properties.
- **Ductility:** ABS has a low elongation at break (5–60%), which makes it prone to brittle failure under impact. If dropped, it can fracture into multiple sharp fragments, posing a risk of cuts. In addition, small broken pieces can become choking hazards if ingested by children.
- **Environmental Concerns:** ABS production has significant environmental implications, with an embodied energy range of 88.9 - 98.3 MJ/kg and a CO₂ footprint range of 3.41-3.77 kg/kg.

4.3.2 Motor

The Russell Hobbs Hand Mixer utilizes a 125 W brushed motor. The motor comprises components numbered 21 through 29 in the bill of materials. The total mass of the motor components is approximately 313 g, with the stator contributing the most weight (133 g). The following advantages and disadvantages compare the brushed motor used in this design to its alternative, a brushless motor.

Advantages

- **Cost-Effective:** Brushed motors are generally less expensive than brushless motors.

Disadvantages

- **Lower efficiency:** Brushed motors are less efficient (80% energy efficiency) than brushless designs (90% energy efficiency) [10], this means that more energy is converted to heat and sound. The brushes rubbing against the commutator generate friction and heat, degrading motor performance and reducing lifespan.
- **Weight:** One notable disadvantage of the brushed motor is its weight compared to brushless alternatives. The necessity of the commutator and brushes to transfer current to the rotor adds weight [10]. More weight means the mixer is harder to hold for extended periods, making it less ergonomic and more challenging for a child to handle comfortably.
- **Power and Heat Generation:** Customer reviews highlighted issues with the motor being excessively fast, even at the lowest speed setting, and prone to overheating [1]. This is likely due to the high power output of the brushed motor combined with inefficient heat dissipation.

4.3.3 Injection Moulding

The primary manufacturing method for the casing and all plastic internal components was injection moulding. This process was likely chosen because it can produce complex shapes with fine detail while requiring little additional finishing. It is also highly efficient for large-scale production, with the capability to manufacture between 60 and 1,000 units per hour, making it a cost-effective option for factory settings. In addition, this method allows for consistent quality and repeatability, which is essential for mass manufacturing.

Although overmoulding, a variation of injection moulding, is not currently used in this design, it could be a valuable addition to improve child-friendliness. Overmoulding enables the integration of multiple materials within a single component, allowing soft-touch grips to enhance handling and vibrant colours for visual appeal.

5 Redesign

This section focuses on improving key components of the Russell Hobbs Hand Mixer to align it more closely to the Krusty Kompany's focus of playful, kid-friendly designs. The main casing and the motor have been redesigned.

5.1 Main Casing Improvements

For the redesign of the main casing, it has been reimagined to be child-friendly aesthetic and have a playful aesthetic. This redesign focuses on two key aspects:

- **Optimizing perceived softness:** The casing material was selected to provide a pleasant tactile experience for children, encouraging them to interact with the mixer under the supervision of an adult. A soft feel would make the appliance more approachable and less intimidating for young users.
- **Maintaining strength while being lightweight:** The casing material should be durable enough to withstand accidental drops without breaking, yet light enough for children to easily handle.
- **Vibrant Colours:** The casing material should be chosen to allow easy dyeing or pigmentation, ensuring a bright and visually appealing design for children.

To achieve these goals, an Ashby plot was created using Granta EduPack (Figure 3). The plot incorporates two key material indices:

- **Perceived softness (S):** Based on a study by Hui Niu at the University of Warwick, perceived softness strongly correlates with Young's modulus (E) and Vickers hardness (H). Lower values of both properties indicate a softer feel. Perceived softness can be approximated as:

$$S = EH$$

- **Specific strength:** Calculated as tensile strength divided by density, this index helps identify materials that are both strong and lightweight.

Young's modulus also indicates how rigid a material is. The casing material should be rigid enough to maintain its shape during regular use but flexible enough to deform rather than shatter upon impact.

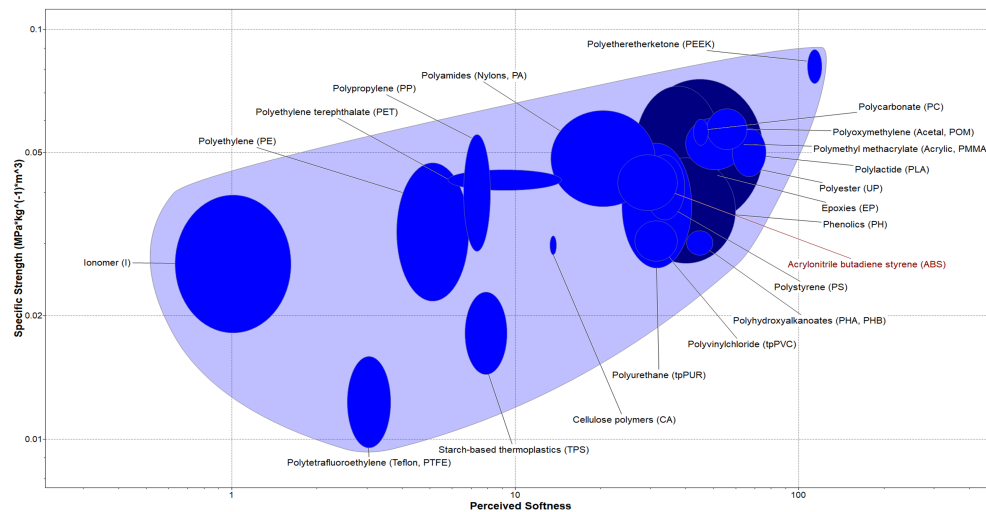


Figure 3: Ashby Plot comparing Perceived Softness and Specific Strength for main casing improvements

Polypropylene (PP) was selected over the original ABS casing due to its improved balance of perceived softness and specific strength while still being reasonably rigid.

Polypropylene offers a specific strength of approximately $0.0550 \text{ Mpa} \cdot \text{kg}^{-1} \cdot \text{m}^3$, which is higher than ABS's $0.0489 \text{ Mpa} \cdot \text{kg}^{-1} \cdot \text{m}^3$. Furthermore, PP perceived softness is significantly improved due to its lower Young's modulus (0.824 - 1.02 GPa) and Vickers hardness (8 HV), making it much softer to hold compared to ABS (Young's modulus 2.07 - 2.76 GPa and 10 - 15 HV).

Advantages of Polypropylene:

1. Safety and Ergonomics:

- The lower Young's modulus (0.824-1.02 GPa) and Vickers' hardness (8 HV) provide a softer tactile experience, making it more comfortable for children to hold.
- Higher elongation (112-483% versus ABS 5-60%) allows the casing to deform rather than crack upon impact, reducing the risk of sharp fragmentation.

2. Household Suitability:

- The lower density ($895\text{-}909 \text{ kg} / \text{m}^3$ vs. $\text{ABS } 1030\text{-}1060 \text{ kg} / \text{m}^3$) results in a lighter hand mixer, improving ease of use.
- High resistance to moisture and chemicals for long-lasting durability in kitchen environments.

3. Manufacturability:

- The lower embodied energy in primary production (64.6 MJ/kg for PP vs. 88.9 MJ/kg for ABS) results in a reduced overall energy consumption.
- Supports vibrant pigmentation for a playful aesthetic.

5.2 Grip Improvement

To enhance the child-friendly design of the hand mixer, the grip areas should be thermally insulative, soft to the touch, and grippy. Young users have sensitive hands and may experience discomfort from prolonged use. A thermally insulating grip prevents heat transfer from the motor for comfort and safety. A soft touch grip improves ergonomics, reducing fatigue, while a high coefficient of friction prevents slipping, improving control even with wet or greasy hands. This can be achieved through overmoulding, where a secondary material is moulded over the polypropylene casing for improved functionality. The overmoulded material must have low thermal conductivity, low perceived softness, support vibrant colours, and have a high coefficient of friction. These overmoulded pieces will be placed on parts of the casing where a child typically holds the mixer, not only the handle.

An Ashby plot (Figure 4) was created with perceived softness on the x-axis and thermal conductivity on the y-axis. The ideal material lies in the lower left quadrant, indicating low values for both properties.

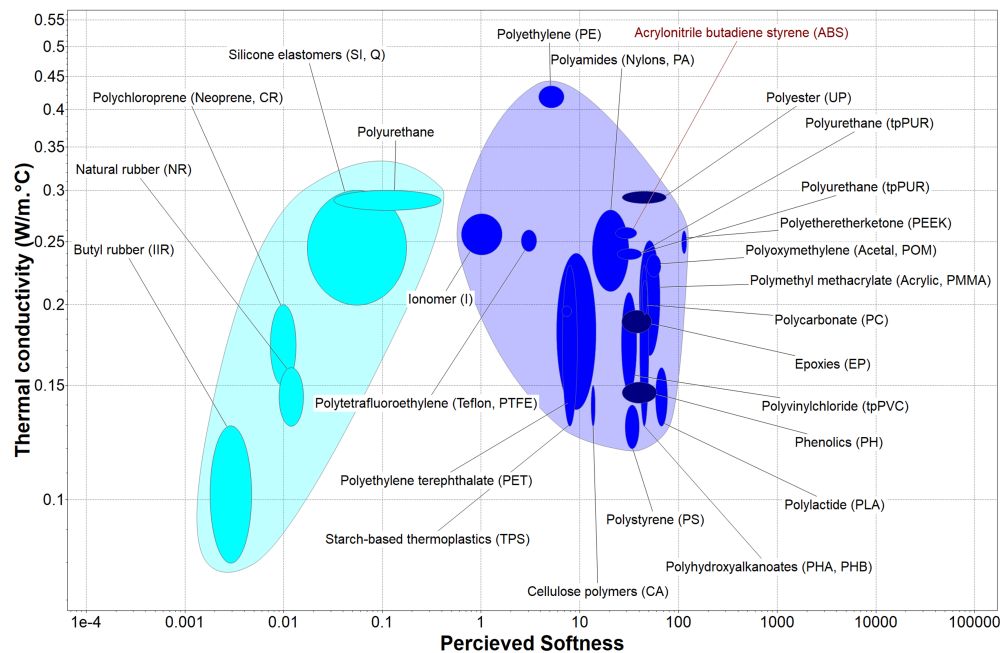


Figure 4: Ashby Plot comparing Perceived Softness and Thermal Conductivity for grip improvements

Following the analysis of the Ashby plot, butyl rubber was identified as the optimal material for the grip sections of the hand mixer. Fortunately, butyl rubber can be effectively overmoulded onto polypropylene. This material exhibited the lowest perceived softness and thermal conductivity among all polymers and elastomers.

Butyl rubber has several advantageous properties. It has a thermal conductivity of 0.13 W/m·°C, lower than the polypropylene casing at 0.199 W/m·°C and the original ABS material at 0.263 W/m·°C. Furthermore, butyl rubber has a high static coefficient of friction [11], which is higher than both polypropylene and ABS, which ensures excellent grip for children. It can also be pigmented like polypropylene, allowing for vibrant and playful colours that appeal to children.

5.3 Motor Improvements

The original hand mixer utilized a brushed motor, which has been replaced with a brushless motor in the redesigned version. Brushless motors operate at an approximate efficiency of 90% [10], significantly reducing heat generation due to friction. In addition, brushless motors produce less noise, making the appliance quieter. A quieter appliance reduces sensory discomfort, making it less intimidating for children and more comfortable to use. The motor wattage will also be reduced from 125 W to 75 W. The decision to lower the power output addresses concerns that the original motor was too powerful and fast, potentially overwhelming child users. In addition, reduced power consumption further minimizes heat production, reducing the risk of overheating. These changes result in a more child-friendly appliance that is safer, more efficient and better suited for its intended use.

5.4 Eco-Audit

To re-evaluate the CO₂ and the embodied energy of the redesigned hand mixer, Granta Edupack [3] was used again. The comparison graphs are shown in Figure 11 and Figure 12 in Appendix F. Overall, CO₂ was reduced by 14% and the embodied energy was reduced by 18%. The greatest reduction was in usage, 40% lower in both CO₂ and embodied energy. This was because the motor was 75 W instead of the original 125 W. The materials also reduced in CO₂ and slightly in embodied energy, this was only due to the outer shell and control panel being changed from ABS to PP.

5.5 Semantic Differential Scales

A large aspect of Krusty Kompany's design goals is to appeal to children and be playful. To measure this, a survey was conducted with eight participants, who were asked to evaluate the original and redesigned materials on a scale of 1-5 in different factors related to child friendliness. Each participant was provided with a description of each material before answering.

- **Unsafe - Safe:** The product should not cause injury during use.
- **Uncomfortable - Comfortable:** The product should be easy for children to hold and use.
- **Heavy - Lightweight:** Lighter products are easier to handle, especially for children.
- **Serious - Playful:** A playful design encourages engagement and enjoyment.
- **Fragile - Durable:** Durability ensures the product can withstand rough handling.

The results of the survey were formed into a semantic differential scale, shown in Figure 5. The survey was used throughout the redesign to confirm that the materials were perceived as safe and kid-friendly.

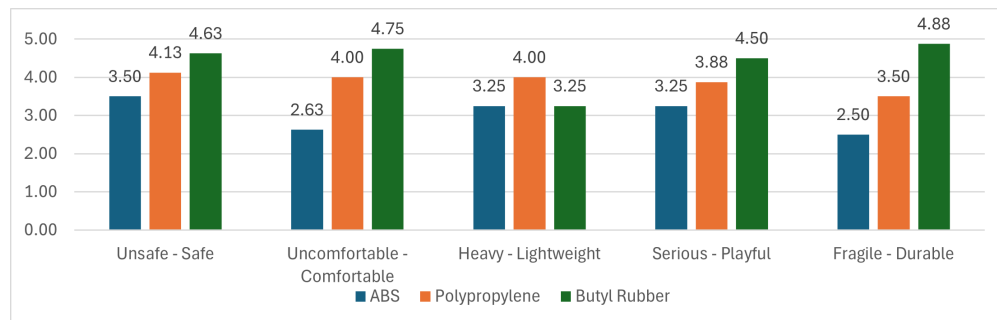


Figure 5: Bar chart comparing materials on a 1-5 scale across factors: safety, comfort, weight, playfulness, and durability

6 Conclusion

In summary, the analysis indicated that the materials had the most significant impact on the embodied energy and footprint of the Russell Hobbs Food Collection Hand Mixer's embodied energy and CO₂ footprint, with ABS being the largest contributor due to its high process energy (88.9–98.3 MJ/kg) and significant mass (27% of the product). The original budget-friendly design relied on cheaper materials and a brushed motor, which led to overheating issues.

The redesign of Krusty Kompany prioritized a playful and child-friendly approach. The casing was switched to polypropylene, making it softer, lighter, and more impact resistant for improved ergonomics and durability. Butyl rubber grips were added in key areas to improve comfort, thermal insulation, and slip resistance. The exterior was updated with more vibrant colours to increase visual appeal and engagement for children. Furthermore, the 125 W brushed motor was replaced with a 75 W brushless alternative, improving efficiency (90% vs. 80%), reducing the risk of overheating and adjusting the power output to make it more manageable for young users. The lighter motor also improves handling. These changes improve safety, usability, and sustainability by lowering the embodied energy and CO₂ emissions, particularly in the usage. The redesigned mixer aligns with Krusty Kompany's goals of creating a safe, child-friendly, and playful product.

7 References

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Appendices

A Teardown Diagram

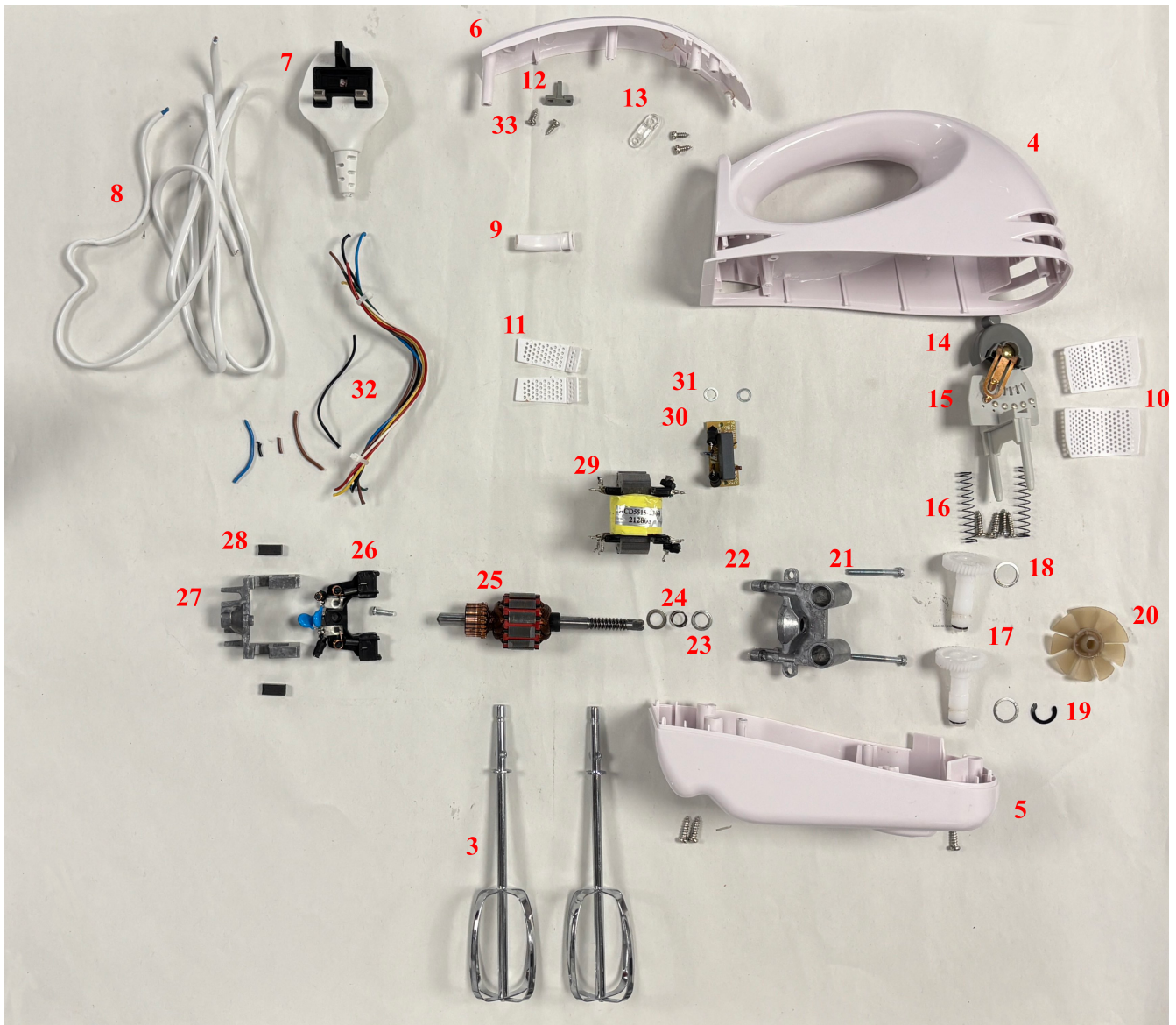


Figure 6: Tear-down of the Russell Hobbs Hand Mixer, illustrating all internal components.

B Teardown Metadata

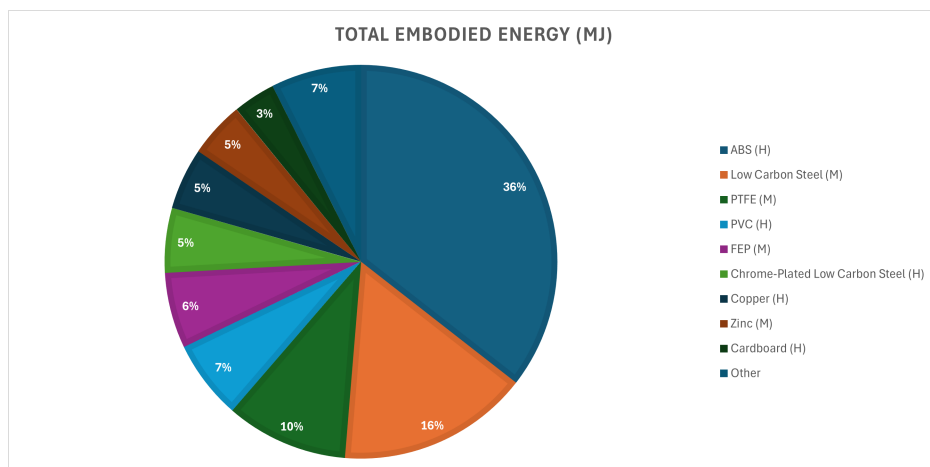


Figure 7: Metadata of the Russell Hobbs Hand Mixer, showing the split of embodied energy per material.

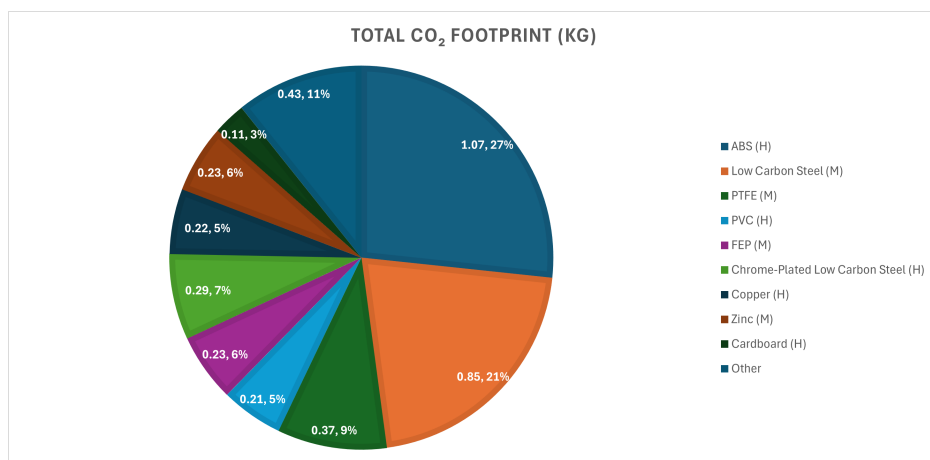


Figure 8: Metadata of the Russell Hobbs Hand Mixer, showing the split of CO₂ per material.

C Bill Of Materials

Table 3: Bill of Materials with Embodied Energy and CO₂ Footprint with Confidence Levels (C, EE, G)

Part No.	Quantity	Name	Material	Mass (kg)	Manufacturing Method	Embodied Energy (MJ)	CO ₂ Footprint (kg)
1	1	Packaging Box	Cardboard (C)	0.081	Die Cutting and Printing (C)	2.30	0.11
2	2	Packaging Bag	LDPE (C)	0.008	Blown Film Extrusion (C)	0.65	0.02
3	2	Beaters	Chrome-Plated Low Carbon Steel (C)	0.043	Metal Stamping (C)	3.51	0.29
4	1	Shell (Upper)	ABS (C)	0.106	Injection Moulding (C)	12.10	0.54
5	1	Shell (Lower)	ABS (C)	0.084	Injection Moulding (C)	9.59	0.43
6	1	Control Panel	ABS (C)	0.015	Injection Moulding (C)	1.71	0.08
7	1	Plug with Fuse	PVC (C)	0.054	Injection Moulding (EE)	4.24	0.21
8	1	Mains Cable	Copper (C)	0.036	Wire Drawing (C)	2.81	0.18
9	1	Cable Sleeve	PVC (C)	<0.001	Polymer Extrusion (C)	0.07	0.00
10	2	Large Vent	PE (C)	0.005	Injection Moulding (C)	1.02	0.04
11	2	Small Vent	PE (C)	<0.001	Injection Moulding (C)	0.20	0.01
12	1	T-Peg	ABS (C)	<0.001	Injection Moulding (C)	0.11	0.01
13	1	Clear Bracket	PMMA (C)	<0.001	Injection Moulding (C)	0.13	0.01
14	1	Speed Lever	PTFE (EE)	0.009	Injection Moulding (C)	2.85	0.16
15	1	Dial Contact Holder	FEP (EE)	0.01	Injection Moulding (C)	3.17	0.18
16	2	Spring	Spring Steel (C)	<0.001	Cold Rolling and Heat Treatment (C)	0.08	0.01
17	2	Gear with Shaft	PTFE (EE)	0.006	Injection Moulding (C)	3.80	0.21
18	2	M11 Washers	Stainless Steel (C)	<0.001	Stamping (C)	0.17	0.01
19	2	Circlips	Spring Steel (C)	<0.001	Stamping (EE)	0.08	0.01
20	1	Cooling Fan	FEP (EE)	0.003	Injection Moulding (C)	0.95	0.05
21	2	Long Screws	Low Carbon Steel (EE)	0.002	Thread Rolling (C)	0.13	0.01
22	1	Beater Bracket	Zinc (EE)	0.028	Die Casting (EE)	1.61	0.12
23	2	M7 Washer	Stainless Steel (C)	<0.001	Stamping (EE)	0.17	0.01
24	1	Small Spring	Spring Steel (C)	<0.001	Cold Rolling and Heat Treatment (C)	0.04	0.00
25	1	Rotor	Low Carbon Steel (EE)	0.111	Stamping and Winding (C)	4.52	0.37
26	1	Graphite Holder & Springs, Diodes, Capacitors	PPQ (G)	0.01	Injection Moulding (C)	0.89	0.04
27	1	Rotor Bracket	Zinc (EE)	0.025	Die Casting (C)	1.43	0.11
28	2	Graphite Contacts	Graphite (C)	<0.001	Moulding (EE)	0.08	0.01
29	1	Stator	Low Carbon Steel (EE)	0.133	Stamping (C)	5.42	0.44
30	1	Capacitor (MKP-X2)	Polypropylene (C)	0.008	Film Winding (C)	0.59	0.03
31	2	M5 Washers	Stainless Steel (C)	<0.001	Stamping (C)	0.17	0.01
32	1	Internal Wiring	Copper (C)	0.007	Wire Drawing (C)	0.55	0.04
33	9	Phillips Head Screw	Low Carbon Steel (EE)	<0.001	Thread Rolling (C)	0.30	0.02
34	3	TX15 Screw	Low Carbon Steel (EE)	0.0007	Thread Rolling (C)	0.07	0.01

D Plastics Identification Flowchart

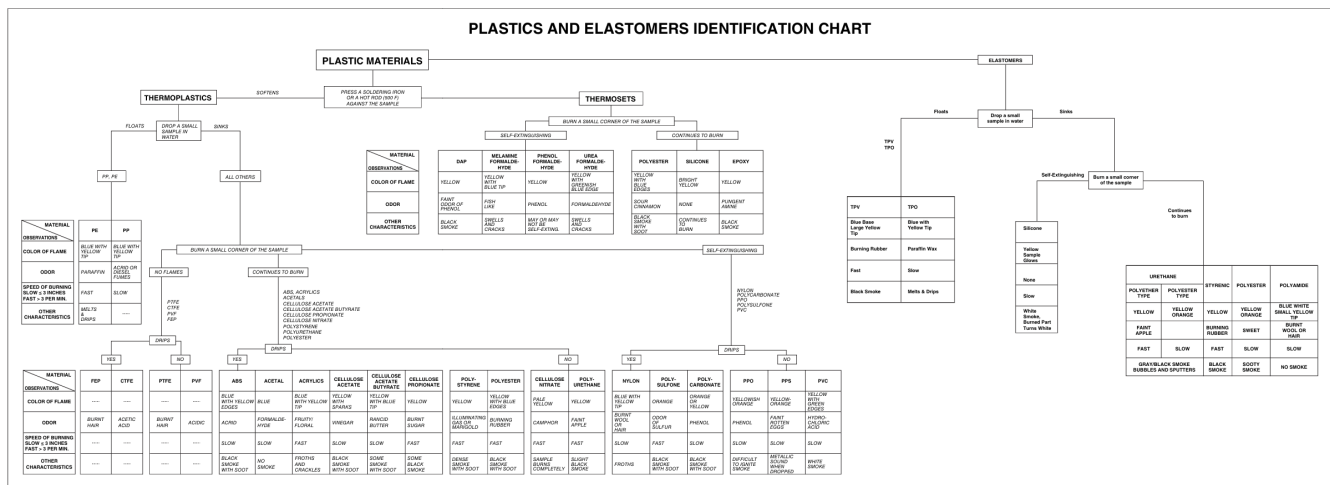


Figure 9: Plastic Identification Flowchart [12]

E Eco-Audit Graph

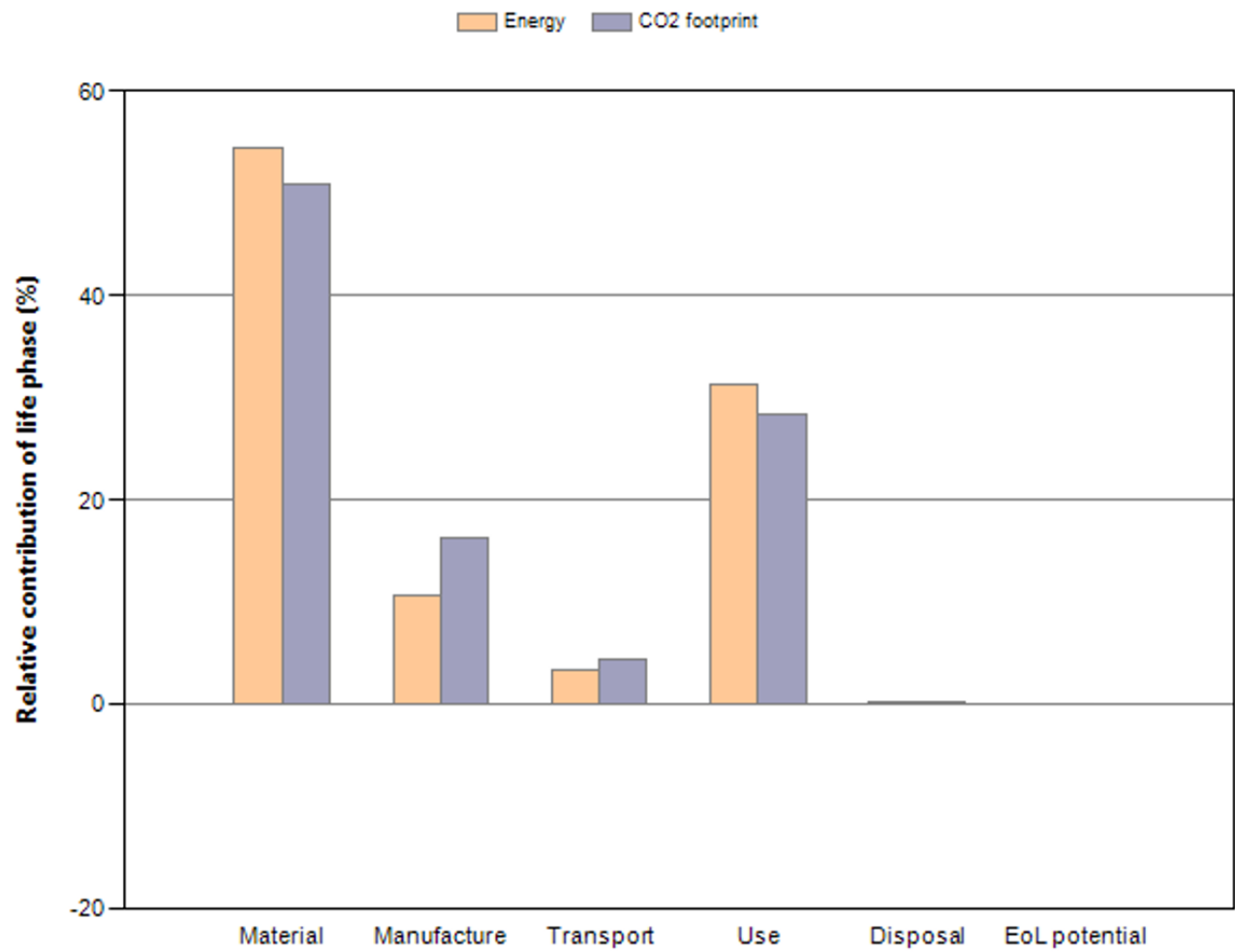


Figure 10: Graph of the Eco-Audit of Russell Hobbs Hand Mixer across the phases[3]

F Re-Evaluated Eco-Audit Graph

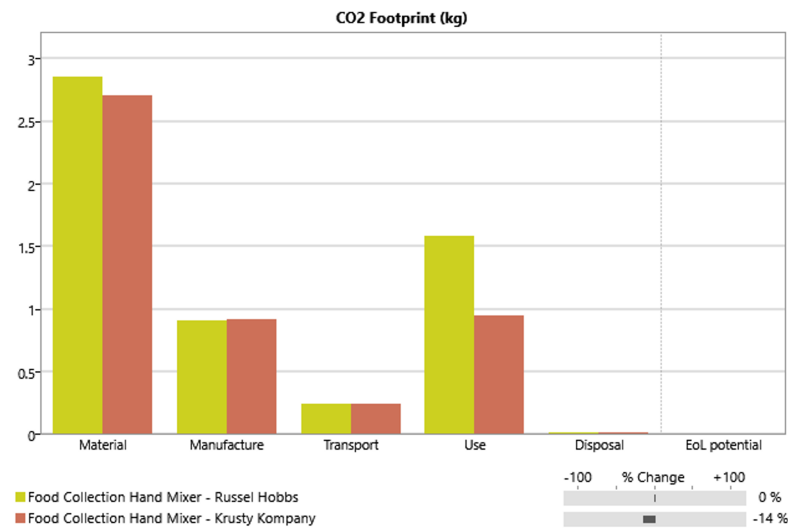


Figure 11: Comparison Graph of Original and Redesign for CO₂ [3]

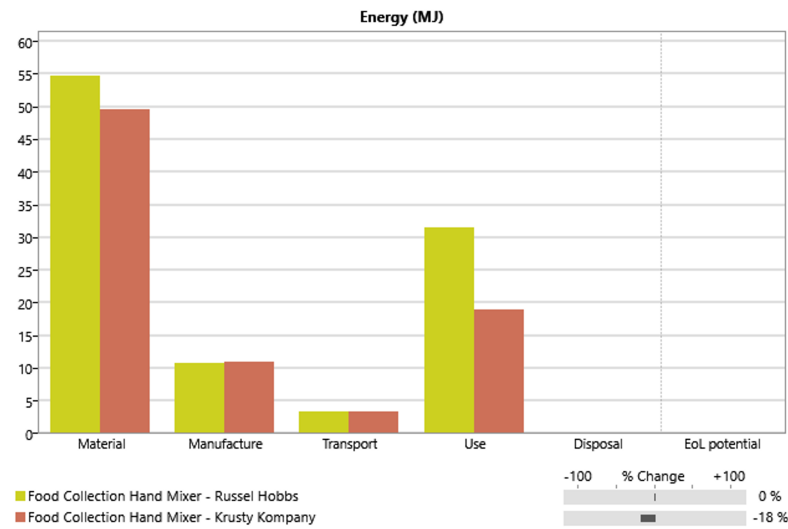


Figure 12: Comparison Graph of Original and Redesign for Embodied Energy [3]